THREE OPEN PROBLEMS OF HU AND VIRTANEN ON THE BERGER–COBURN PHENOMENON

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ABSTRACT. For a bounded symbol f, the Berger–Coburn phenomenon for an operator ideal $\mathcal{M}$ is the statement that the Hankel operator H_f belongs to $\mathcal{M}$ if and only if $H_{\bar{f}}$ does. Hu and Virtanen asked whether there are nontrivial operator ideals with this property on the Hardy space $H^2(\mathbb{T})$ or on the Bergman space $A^2(\mathbb{B}_n)$, for which p the phenomenon holds for the Schatten classes S_p on doubling Fock spaces, and whether it holds for compact Hankel operators on doubling Fock spaces. We show that on $H^2(\mathbb{T})$ and $A^2(\mathbb{B}_n)$ the only such operator ideal is the whole space of bounded operators; that on doubling Fock spaces the phenomenon holds for S_p on every space if and only if $p = 2$, and on some space if and only if $p > 1$; and that it holds for compact Hankel operators on every doubling Fock space. The second result also answers two questions of Asghari, Hu and Virtanen.

1. INTRODUCTION

Let H be a Hilbert space, let $X = P(H)$ be the range of an orthogonal projection P, and for a bounded operator A on H let $H_A = (I-P)A\colon X \to H$ be the corresponding abstract Hankel operator. Following Hu and Virtanen [7, Definition 1], the *Berger–Coburn phenomenon* (BCP) holds for an operator ideal $\mathcal{M}$ in $B(X, H)$ and a class $\mathcal{A}$ of operators if $H_A \in \mathcal{M}$ if and only if $H_{A^*} \in \mathcal{M}$ for every $A \in \mathcal{A}$. We are concerned with multiplication operators $A = M_f$, $f \in L^\infty$, so that $A^* = M_{\bar{f}}$ and the BCP says that

$$H_f \in \mathcal{M} \iff H_{\bar{f}} \in \mathcal{M}, \qquad H_f = (I - P)M_f,$$

for every bounded symbol f. Three settings occur. The first two are the Hardy space $H^2 = H^2(\mathbb{T})$ inside $L^2(\mathbb{T})$, with P the Riesz projection, and the Bergman space $A^2 = A^2(\mathbb{B}_n)$ of the unit ball $\mathbb{B}_n \subset \mathbb{C}^n$ inside $L^2(\mathbb{B}_n)$, with P the Bergman projection. The third is a doubling Fock space. Let ϕ be a subharmonic function on $\mathbb{C}$, not identically zero, such that $\Delta\phi \, \mathrm{d}A$ is a doubling measure, and let L^2_ϕ be the space of measurable functions on $\mathbb{C}$ with

$$\|f\|^2_{2,\phi} = \int_{\mathbb{C}} |f(z)|^2 e^{-2\phi(z)} \, \mathrm{d}A(z) < \infty.$$

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The doubling Fock space is $F^2_\phi = L^2_\phi \cap H(\mathbb{C})$, and P is the orthogonal projection of L^2_ϕ onto F^2_ϕ. When $\phi \in C^2(\mathbb{C})$ and $\Delta\phi \simeq 1$, we recover the generalized Fock spaces of [5]; the classical Fock space corresponds to $\phi(z) = |z|^2/2$.

Berger and Coburn [3] proved that the BCP holds for the compact operators on the classical Fock space, and Hu and Virtanen [8, Theorem 1.2] extended this to weights $\phi \in C^2(\mathbb{C}^n)$ whose real Hessian is comparable to the identity. Hu and Virtanen also observed that it fails for the compact operators on H^2 and on $A^2(\mathbb{B}_n)$: a bounded holomorphic symbol f gives $H_f = 0$, while $H_{\bar{f}}$ need not be compact [7, Example 2]. For the Schatten classes S_p on Fock spaces, Bauer [2] settled $p = 2$ on the classical Fock space, and Hu and Virtanen [5,6] proved the BCP for S_p, $1 < p < \infty$, when $\Delta\phi \simeq 1$ [7, Theorem 3]. Xia [12] showed that the symbol

$$f(z) = \begin{cases} 1/z, & |z| \geq 1, \\ 0, & |z| < 1, \end{cases} \tag{1}$$

gives a trace class H_f and a non-trace class $H_{\bar{f}}$ on the classical Fock space, and Hu and Virtanen used the same symbol to show that the BCP fails for S_p, $0 < p \leq 1$, when $\Delta\phi \simeq 1$ [7, Theorem 7]. Asghari, Hu and Virtanen [1] then characterized S_p membership of H_f on doubling Fock spaces in terms of integral distance to holomorphic functions [1, Theorem 1.2], proved the BCP for S_2 on every doubling Fock space [1, Theorem 1.4], and showed that it fails for S_p, $0 < p \leq 1$, whenever the growth exponent β_ϕ of (4) below satisfies $\beta_\phi \leq (1-p)/(1-p/2)$ [1, Theorem 1.6].

Hu and Virtanen formulated the following open problems [7, Sections 2 and 3]. We quote them verbatim, with the numbering of the arXiv version 2304.09716v1 of [7]; for [1] we use the numbering of the arXiv version 2312.06656v2.

Open Problem A (Hu and Virtanen [7, Open Problem 1])**.** *Are there nontrivial operator ideals in $B(H^2(\mathbb{T}), L^2(\mathbb{T}))$ or in $B(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$ for which the Berger–Coburn phenomenon holds?*

Open Problem B (Hu and Virtanen [7, Open Problem 2])**.** *Let F^2_ϕ be a doubling Fock space. For which values of p does the Berger–Coburn phenomenon hold for $S_p(F^2_\phi, L^2_\phi)$?*

Open Problem C (Hu and Virtanen [7, Open Problem 3])**.** *Let F^2_ϕ be a doubling Fock space and $f \in L^\infty$. Is it true that if $H_f \colon F^2_\phi \to L^2_\phi$ compact, then $H_{\bar{f}}$ is also compact?*

In [7] the last two problems are introduced as the question of "whether the statements of Theorems 3 and 7 and the theorem of Berger and Coburn remain true for doubling Fock spaces". Asghari, Hu and Virtanen restate Open Problem B as two questions [1, Open Problems 1.5 and 1.8]: "Does the Berger–Coburn phenomenon hold true for other Schatten classes S_p when

$1 < p < \infty$?" and "Determine whether the Berger–Coburn phenomenon fails for $S_p(F^2_\phi, L^2_\phi)$ when $0 < p \le 1$ and $\Delta\phi \, \mathrm{d}A$ is doubling."

The main purpose of the present paper is to answer these questions. The term operator ideal is not specified further in [7]; we use only the following property, which every operator ideal in the usual sense has. A set $\mathcal{M} \subset B(X, H)$ is called an *ideal* if it is nonempty and $ATB \in \mathcal{M}$ whenever $T \in \mathcal{M}$, $A \in B(H)$ and $B \in B(X)$.

Theorem 1. *Suppose that $X = H^2(\mathbb{T}) \subset L^2(\mathbb{T})$, or that $X = A^2(\mathbb{B}_n) \subset L^2(\mathbb{B}_n)$ for some $n \ge 1$, and that $\mathcal{M}$ is an ideal in $B(X, L^2)$. Then the BCP holds for $\mathcal{M}$ if and only if $\mathcal{M} = B(X, L^2)$.*

The answer to Open Problem A is therefore negative, whichever reading of *nontrivial* is used: neither $\{0\}$ nor any proper ideal has the BCP. On doubling Fock spaces the answer to Open Problem B depends on p in the following way.

Theorem 2. *Suppose that $0 < p < \infty$.*

(a) *If $0 < p \le 1$, then the BCP fails for $S_p(F^2_\phi, L^2_\phi)$ on every doubling Fock space F^2_ϕ.*
(b) *If $1 < p < 2$ and $m \ge 2p/(2-p)$, then the BCP fails for $S_p(F^2_\phi, L^2_\phi)$ when $\phi(z) = |z|^m$.*
(c) *If $2 < p < \infty$ and $N > 1/(p-2)$ is an integer, then the BCP fails for $S_p(F^2_\phi, L^2_\phi)$ when $\phi(x+iy) = y^{2N+2}$.*

In each case the failure is witnessed by a symbol $f \in L^\infty$ with $H_f \in S_p$ and $H_{\bar f} \notin S_p$.

Combining Theorem 2 with [1, Theorem 1.4] for $p = 2$ (stated there for smooth ϕ; Lemma 3 (a) below removes this assumption) and [7, Theorem 3] for the classical Fock space, we obtain the following answer.

Corollary 1. *Suppose that $0 < p < \infty$. The BCP holds for $S_p(F^2_\phi, L^2_\phi)$ on every doubling Fock space if and only if $p = 2$, and on at least one doubling Fock space if and only if $p > 1$.*

In particular, Theorem 7 of [7] remains true for doubling Fock spaces, which answers [1, Open Problem 1.8], while Theorem 3 of [7] does not remain true for any $p \ne 2$, which answers [1, Open Problem 1.5] in the negative. Part (a) removes the condition on β_ϕ in [1, Theorem 1.6]. Part (b) is the counterpart for $1 < p < 2$ of [1, Corollary 1.7], which gives failure for $0 < p \le 1$ and $\phi(z) = |z|^m$ under the same condition $m \ge p/(1-p/2) = 2p/(2-p)$. The answer to Open Problem C is affirmative.

Theorem 3. *Suppose that F^2_ϕ is a doubling Fock space and that $f \in L^\infty$. Then $H_f \colon F^2_\phi \to L^2_\phi$ is compact if and only if $H_{\bar f}$ is compact.*

To the best of our knowledge, Theorems 1 to 3 are new.

Let us briefly describe our approach. Theorem 1 rests on the observation in [7, Example 2]: an ideal with the BCP contains $H_f = 0$ for every bounded

holomorphic f, hence it contains every $H_{\bar{f}}$. A Calkin-type argument shows that an ideal containing one noncompact operator is everything, and the explicit symbol $(1-z_1)^i$ gives a noncompact $H_{\bar{f}}$ with a two-line computation. For doubling Fock spaces everything goes through the characterization of Asghari, Hu and Virtanen, $H_f \in S_p$ if and only if $f \in \mathrm{IDA}^{p,2,-2/p}$. We use two local estimates for the distance $G_r(f)$ to holomorphic functions: $G_r(f)$ is at most $\rho|\bar{\partial} f|$ (by a local Cauchy transform), and $G_r(\bar{f}) = \mathrm{MO}_r(f)$ is at least a constant times $\rho|f'|$ when f is holomorphic. Thus Open Problem B becomes a comparison of

$$\int_{\mathbb{C}} \rho^{p-2}|\bar{\partial} f|^p \,\mathrm{d}A \qquad \text{and} \qquad \int_{\mathbb{C}} \rho^{p-2}|\partial f|^p \,\mathrm{d}A,$$

in agreement with [1, Remark 6.1]. For $p = 2$ the weight ρ^{p-2} is 1, and the two integrals are equal for smooth compactly supported f by integration by parts; for $p \neq 2$ we choose ϕ so that ρ^{p-2} is small where $\bar{\partial} f$ lives and large where ∂f lives. For $0 < p \leq 1$ we show that every doubling weight has "flat spots", large disks on which ρ is nearly constant, and we place poles of f there. Theorem 3 uses the compactness hypothesis on the normalized reproducing kernels: $G_r(f)(z) \to 0$ as $z \to \infty$. Rescaling at scale $\rho(z)$, the symbol becomes locally close to holomorphic functions which are bounded, and Liouville's theorem forces them to be constant. This gives vanishing mean oscillation, which is symmetric in f and $\bar{f}$; this is the mechanism Hu and Virtanen describe for the Hagger–Virtanen characterization in [7, Section 2].

The present paper is organized into five additional sections. In Section 2 we prove Theorem 1. In Section 3 we collect the facts on doubling Fock spaces that we need and prove the two local estimates. In Section 4 we prove Theorem 2, one part at a time. In Section 5 we prove Theorem 3. Section 6 contains concluding remarks, including the question left open by Corollary 1: the set of p for which the BCP holds on a given doubling Fock space.

2. Operator ideals on the Hardy and Bergman spaces

In this section X is either H^2 or $A^2(\mathbb{B}_n)$, L^2 is $L^2(\mathbb{T})$ with normalized arc length m or $L^2(\mathbb{B}_n)$ with normalized volume measure v, and P is the orthogonal projection of L^2 onto X. The proof of Theorem 1 has two steps. The first is a Calkin-type lemma, which shows that the BCP is very costly for an ideal: one noncompact operator in it is enough to make it everything.

Lemma 1. *Suppose that H_1 and H_2 are separable Hilbert spaces, H_1 infinite dimensional, and that $\mathcal{M} \subset B(H_1, H_2)$ satisfies $ATB \in \mathcal{M}$ whenever $T \in \mathcal{M}$, $A \in B(H_2)$ and $B \in B(H_1)$. If $\mathcal{M}$ contains a noncompact operator, then $\mathcal{M} = B(H_1, H_2)$.*

Proof. Let $T \in \mathcal{M}$ be noncompact. For $\varepsilon > 0$ let E_ε be the spectral projection of T^*T for the set $\{\lambda \geq \varepsilon^2\}$. Then $\|T^*T - T^*TE_\varepsilon\| \leq \varepsilon^2$. If every E_ε had finite rank, T^*T would be a norm limit of finite rank operators,

hence compact, and so would $T = U(T^*T)^{1/2}$ (polar decomposition). We may therefore fix $\varepsilon > 0$ such that $E_\varepsilon H_1$ is infinite dimensional.

Since H_1 is separable, there is an isometry $V \in B(H_1)$ with range $E_\varepsilon H_1$. For every $x \in H_1$ we have $\|TVx\|^2 = \langle T^*TVx, Vx\rangle \geq \varepsilon^2\|x\|^2$. Hence TV is injective with closed range, and the operator $L \in B(H_2, H_1)$ defined as $(TV)^{-1}$ on the range of TV and as 0 on its orthogonal complement satisfies $LTV = I$ and $\|L\| \leq 1/\varepsilon$. For any $S \in B(H_1, H_2)$ we can now write

$$S = (SL)\,T\,V, \qquad SL \in B(H_2), \quad V \in B(H_1),$$

so $S \in \mathcal{M}$. □

The second step is a bounded holomorphic symbol f for which $H_{\bar f}$ is not compact. Hu and Virtanen use a Blaschke product, Wolff's theorem on $H^\infty \setminus \mathrm{VMO}$, and Axler's and Li's characterizations of compact Hankel operators with conjugate analytic symbols [7, Example 2]. The following explicit symbol handles both settings with one computation. Its key property is that $g(\zeta) = (1-\zeta)^i$ is bounded but oscillates infinitely often as $\zeta \to 1$, so its Bloch seminorm $(1-|\zeta|^2)|g'(\zeta)|$ does not tend to 0.

Lemma 2. *Let $g(\zeta) = (1-\zeta)^i = \exp(i\log(1-\zeta))$ for $\zeta \in \mathbb{D}$, with the principal branch of the logarithm. Let f be the boundary function of g on $\mathbb{T}$ in the Hardy case, and $f(z) = g(z_1)$ on $\mathbb{B}_n$ in the Bergman case. Then $f \in L^\infty$, $H_f = 0$, and $H_{\bar f}$ is not compact.*

Proof. Since $\mathrm{Re}(1-\zeta) > 0$ on $\mathbb{D}$, we have $|g(\zeta)| = \exp(-\arg(1-\zeta)) \leq e^{\pi/2}$. Hence $g \in H^\infty(\mathbb{D})$, and f is a bounded holomorphic function on $\mathbb{B}_n$, respectively the boundary function of one on $\mathbb{T}$. Thus M_f maps X into X, and $H_f = 0$. Moreover $g'(\zeta) = -i(1-\zeta)^i/(1-\zeta)$, and $|(1-t)^i| = 1$ for $0 < t < 1$, so

$$(1-t^2)|g'(t)| = 1 + t \geq 1, \qquad 0 < t < 1. \tag{2}$$

Let K_a be the reproducing kernel of X at a and $k_a = K_a/\|K_a\|$. We first compute $H_{\bar f}k_a$. For $h \in X$ the function fh lies in X, so

$$\langle \bar f k_a, h\rangle = \langle k_a, fh\rangle = \frac{\overline{f(a)h(a)}}{\|K_a\|} = \langle \overline{f(a)}k_a, h\rangle.$$

Hence $P(\bar f k_a) = \overline{f(a)}k_a$, and $H_{\bar f}k_a = (\bar f - \overline{f(a)})k_a$, that is,

$$\|H_{\bar f}k_a\|^2 = \int |f - f(a)|^2 |k_a|^2. \tag{3}$$

Take $a = t$ in the Hardy case and $a = (t, 0, \ldots, 0)$ in the Bergman case, with $0 < t < 1$, and let σ_a be the involutive automorphism of $\mathbb{D}$ or $\mathbb{B}_n$ which interchanges 0 and a. In both cases the first coordinate of $\sigma_a(z)$ is $(t - z_1)/(1 - tz_1)$ [11, Section 2.2.1]. The normalized kernels are $k_a(z) = (1-t^2)^{1/2}/(1-tz)$ on $\mathbb{T}$ and $k_a(z) = (1-t^2)^{(n+1)/2}/(1-tz_1)^{n+1}$ on $\mathbb{B}_n$, and $|k_a|^2$ is the Jacobian of σ_a with respect to m, respectively v; on

$\mathbb{T}$ this is $|k_a|^2 = |\sigma_a'|$, and on $\mathbb{B}_n$ it is [11, Theorem 2.2.6]. Since σ_a is an involution, the change of variables $z = \sigma_a(\zeta)$ in (3) gives

$$\|H_{\bar f} k_a\|^2 = \|u\|^2, \qquad u = f \circ \sigma_a - f(a) \in X.$$

In words: the Hankel operator applied to the kernel at a measures how far f moves under the automorphism which brings a to the origin. The monomials are orthogonal in X, with $\|z_1\|^2 = 1$ in H^2 and $\|z_1\|^2 = 1/(n+1)$ in $A^2(\mathbb{B}_n)$ [11, Proposition 1.4.9]. Hence $\langle u, z_1\rangle = \|z_1\|^2 \partial_1 u(0)$, and by the chain rule $\partial_1 u(0) = -(1-t^2)g'(t)$. Bessel's inequality and (2) give

$$\|H_{\bar f} k_a\|^2 \geq \frac{|\langle u, z_1\rangle|^2}{\|z_1\|^2} = \|z_1\|^2 (1-t^2)^2 |g'(t)|^2 \geq \frac{1}{n+1},$$

with $n = 1$ in the Hardy case. On the other hand, $k_a \to 0$ weakly as $t \to 1$. Indeed, $\|k_a\| = 1$, polynomials are dense in X, and for a polynomial q we have $\langle q, k_a\rangle = \overline{q(a)}/\|K_a\| \to 0$, because $\|K_a\|^2 = (1-t^2)^{-1}$, respectively $(1-t^2)^{-(n+1)}$. A compact operator maps weakly null sequences to norm null sequences, so $H_{\bar f}$ is not compact. □

Proof of Theorem 1. The BCP clearly holds for $\mathcal{M} = B(X, L^2)$. Conversely, suppose that it holds for an ideal $\mathcal{M}$. Taking $A = 0$ in the definition of an ideal shows that $0 \in \mathcal{M}$. With f as in Lemma 2, we have $H_f = 0 \in \mathcal{M}$, hence $H_{\bar f} \in \mathcal{M}$, and $H_{\bar f}$ is not compact. Since X and L^2 are separable and X is infinite dimensional, Lemma 1 gives $\mathcal{M} = B(X, L^2)$. □

The proof shows more generally that any ideal with the BCP contains $H_{\bar f}$ for every $f \in H^\infty$; this is the obstruction. On Fock spaces it disappears, because bounded entire functions are constant [7, Section 2].

3. Preliminaries on doubling Fock spaces

From now on ϕ is subharmonic on $\mathbb{C}$, not identically zero, and $\mu = \Delta\phi \, \mathrm{d}A$ is doubling. We use the notation of [1]. Let $\rho(z)$ be the radius with $\mu(D(z, \rho(z))) = 1$, and put $D^r(z) = D(z, r\rho(z))$. For $f \in L^2_{\mathrm{loc}}$ and $r > 0$ let

$$G_r(f)(z) = \inf_{h \in H(D^r(z))} \left(\frac{1}{|D^r(z)|} \int_{D^r(z)} |f - h|^2 \, \mathrm{d}A \right)^{1/2},$$

which is $G_{2,r}(f)$ in [1], and let $\mathrm{MO}_r(f)(z)$ be the same quantity with h restricted to constants, which is the mean oscillation $\mathrm{MO}_{2,r}(f)(z)$ of [1]. Clearly $G_r(f) \leq \mathrm{MO}_r(f)$, and $\mathrm{MO}_r(f) = \mathrm{MO}_r(\bar f)$. We will use the following facts, valid for every such ϕ.

(R1) $|\rho(z) - \rho(w)| \leq |z - w|$ for all $z, w \in \mathbb{C}$ [1, (2.1)].

(R2) For every $s > 0$ there is $c_s \geq 1$ such that $c_s^{-1}\rho(z) \leq \rho(w) \leq c_s \rho(z)$ for $w \in D^s(z)$ [1, Lemma 2.1]. In particular, $\int_{D^s(z)} \rho^{-2} \, \mathrm{d}A \leq \pi s^2 c_s^2$.

(R3) There are $C_0, \eta > 0$ and $0 \leq \beta < 1$ such that

$$C_0^{-1} |z|^{-\eta} \leq \rho(z) \leq C_0 |z|^\beta, \qquad |z| > 1, \tag{4}$$

see [1, (1.17)]; the least such β is β_ϕ. In particular $\rho(z)/|z| \to 0$ as $z \to \infty$.

By a theorem of Marco, Massaneda and Ortega-Cerdà [9, Theorem 14] there is a subharmonic $\psi \in C^\infty(\mathbb{C})$ with $|\psi - \phi|$ bounded, $\Delta\psi \, \mathrm{d}A$ doubling, and $\rho_\psi \simeq \rho_\phi$, where the subscript indicates the weight. Asghari, Hu and Virtanen state their results for smooth weights, after this reduction [1, Section 1]. The next lemma records why the reduction is harmless for our purposes, and fixes the form of their characterization that we use.

Lemma 3. *Suppose that $f \in L^\infty$ and that ψ is as above.*

(a) *H_f is compact, respectively in S_p, on F^2_ϕ if and only if it is so on F^2_ψ.*
(b) *Let $0 < p < \infty$ and $r > 0$. Then $H_f \in S_p(F^2_\phi, L^2_\phi)$ if and only if*

$$\int_{\mathbb{C}} \rho^{-2} G_r(f)^p \, \mathrm{d}A < \infty. \tag{5}$$

Proof. (a) Since $|\psi - \phi|$ is bounded, $L^2_\psi = L^2_\phi$ and $F^2_\psi = F^2_\phi$ as sets, with equivalent norms. Write H^ϕ_f and H^ψ_f for the two Hankel operators. Since $P_\psi(fg) \in F^2_\phi$ is annihilated by $I - P_\phi$, we have

$$H^\phi_f = (I - P_\phi) H^\psi_f, \qquad H^\psi_f = (I - P_\psi) H^\phi_f,$$

where $I - P_\phi$ and $I - P_\psi$ are bounded on the common space L^2_ϕ. Compact operators and S_p are ideals, so (a) follows.

(b) For ϕ replaced by ψ, this is the equivalence of (1) and (2) in [1, Theorem 1.2], since $\|f\|^p_{\mathrm{IDA}^{p,2,-2/p}_r} = \int \rho_\psi^{-2} G^\psi_r(f)^p \, \mathrm{d}A$, and it holds for every $r > 0$. Let $c\rho_\phi \le \rho_\psi \le C\rho_\phi$. If $D' \subset D''$ are concentric disks, then restricting $h \in H(D'')$ to D' shows that the normalized distance from f to $H(D')$ is at most $(|D''|/|D'|)^{1/2}$ times the normalized distance from f to $H(D'')$. Hence $G^\phi_r \le (C/c) G^\psi_{r/c}$ and $G^\psi_r \le (C/c) G^\phi_{Cr}$. Since moreover $\rho_\phi^{-2} \simeq \rho_\psi^{-2}$, the integral (5) for ϕ at radius r is dominated by the corresponding integral for ψ at radius r/c, and the integral for ψ at radius r/C is dominated by (5). Combining this with (a) and the ψ case gives (b). □

Throughout we use $\rho = \rho_\phi$ and Lemma 3 (b). The next lemma contains the two local estimates. Part (a) says that a function which is nearly holomorphic in a disk is close to a holomorphic function there; part (b) says that the conjugate of a holomorphic function is far from holomorphic functions, by an amount proportional to its derivative. Part (b) is where the asymmetry between f and $\bar f$ enters. For holomorphic f the best holomorphic approximant of $\bar f$ is the constant $\overline{f(z)}$, which is why the integral mean oscillation IMO of [1] appears.

Lemma 4. *Let $D = D(z, \delta)$, and write* $\operatorname{dist}(F, D)$ *for the normalized distance* $\inf_{h \in H(D)} (|D|^{-1} \int_D |F - h|^2 \, \mathrm{d}A)^{1/2}$.

(a) *If f is C^1 in a neighbourhood of $\overline{D}$, then* $\operatorname{dist}(f, D) \le 2\delta \sup_D |\bar\partial f|$. *In particular* $G_{1/2}(f)(z) \le \rho(z) \sup_{D^{1/2}(z)} |\bar\partial f|$.

(b) *If f is holomorphic in a neighbourhood of $\overline{D}$, then*

$$\operatorname{dist}(\bar f, D) = \left(\frac{1}{|D|} \int_D |f - f(z)|^2 \, \mathrm{d}A \right)^{1/2} \ge \frac{\delta |f'(z)|}{\sqrt{2}}.$$

In particular, if f is holomorphic in a neighbourhood of $\overline{D^{1/2}(z)}$, then $G_{1/2}(\bar f)(z) = \mathrm{MO}_{1/2}(f)(z) \ge \rho(z)|f'(z)|/(2\sqrt{2})$.

Proof. (a) Let u be the Cauchy transform

$$u(w) = \frac{1}{\pi} \int_D \frac{\bar\partial f(\zeta)}{w - \zeta} \, \mathrm{d}A(\zeta).$$

Since $1/(\pi w)$ is a fundamental solution of $\bar\partial$, we have $\bar\partial u = \bar\partial f$ in D in the sense of distributions, so $f - u$ is holomorphic in D by Weyl's lemma. For $w \in D$, the integral of the radially decreasing function $|w - \zeta|^{-1}$ over D is at most its integral over $D(w, \delta)$, which is $2\pi\delta$. Hence $|u| \le 2\delta \sup_D |\bar\partial f|$ on D, and $\operatorname{dist}(f, D) \le \sup_D |u|$. The second statement is the case $\delta = \rho(z)/2$.

(b) Write $f(w) = \sum_{k \ge 0} a_k (w - z)^k$, so that $a_0 = f(z)$ and $a_1 = f'(z)$; the series converges uniformly on $\overline{D}$. If $h \in H(D) \cap L^2(D)$ then, for $0 < \delta' < \delta$, rotation invariance gives $\int_{D(z,\delta')} (w - z)^k h(w) \, \mathrm{d}A(w) = 0$ for $k \ge 1$, because the integrand expands in powers $(w - z)^j$ with $j \ge 1$. Letting $\delta' \to \delta$, we see that $\bar f - \bar a_0$ is orthogonal to $H(D) \cap L^2(D)$. If $h \in H(D)$ is not in $L^2(D)$, then $\bar f - h \notin L^2(D)$ since f is bounded on D. Hence, by Pythagoras' theorem, the constant $\bar a_0$ is the best approximation, and the orthogonality of the powers $(w - z)^k$ gives

$$\int_D |\bar f - \bar a_0|^2 \, \mathrm{d}A = \sum_{k \ge 1} |a_k|^2 \frac{\pi \delta^{2k+2}}{k + 1} \ge |f'(z)|^2 \frac{\pi \delta^4}{2}.$$

Dividing by $|D| = \pi\delta^2$ gives the inequality. Since the best constant in the definition of $\mathrm{MO}_{1/2}$ is the mean, which is $f(z)$ by the mean value property, we obtain $G_{1/2}(\bar f)(z) = \mathrm{MO}_{1/2}(f)(z)$, and the last statement is the case $\delta = \rho(z)/2$. □

For a bounded symbol f which is smooth, Lemmas 3 and 4 show that $H_f \in S_p$ when $\int \rho^{p-2} \sup_{D^{1/2}} |\bar\partial f|^p \, \mathrm{d}A < \infty$, and that $H_{\bar f} \notin S_p$ when $\int_E \rho^{p-2} |f'|^p \, \mathrm{d}A = \infty$, where E is the set of z such that f is holomorphic near $\overline{D^{1/2}(z)}$. All three constructions in the next section are designed around these two integrals.

4. Schatten class Hankel operators on doubling Fock spaces

We prove the three parts of Theorem 2 in the order (b), (c), (a), from the simplest construction to the most involved. In each case we give a bounded

symbol f, check $H_f \in S_p$ with Lemma 4 (a), and check $H_{\bar f} \notin S_p$ with Lemma 4 (b). We write $G = G_{1/2}$ and use Lemma 3 (b) with $r = 1/2$.

Part (b): Xia's symbol on $F^2_{|z|^m}$. For $1 < p < 2$ the weight ρ^{p-2} is large where ρ is small. When $\phi(z) = |z|^m$ with $m > 2$, ρ decays at infinity, so the weight ρ^{p-2} amplifies the derivative of Xia's symbol (1) far out, while $\bar\partial f$ lives on the unit circle only.

Proof of Theorem 2 (b). Let f be Xia's symbol (1). The measure $\Delta\phi \, \mathrm{d}A$ is doubling, and $\rho(z) \simeq |z|^{1-m/2}$ for large $|z|$ [1, Lemma 2.5]. By (R3) there is $R > 2$ such that $\overline{D^{1/2}(z)}$ does not meet the closed unit disk when $|z| \geq R$. For such z, f is holomorphic near $\overline{D^{1/2}(z)}$, so $G(f)(z) = 0$. For $|z| < R$ we use $h = 0$ in the definition of G to get $G(f)(z) \leq 1$. Since ρ is continuous and positive by (R1),

$$\int_{\mathbb{C}} \rho^{-2} G(f)^p \, \mathrm{d}A \leq \int_{|z|<R} \rho^{-2} \, \mathrm{d}A < \infty,$$

and Lemma 3 gives $H_f \in S_p$. For $|z| \geq R$, Lemma 4 (b) gives $G(\bar f)(z) \geq \rho(z)|z|^{-2}/(2\sqrt{2})$. Hence

$$\begin{aligned} \int_{\mathbb{C}} \rho^{-2} G(\bar f)^p \, \mathrm{d}A &\geq c \int_{|z|\geq R} \rho^{p-2} |z|^{-2p} \, \mathrm{d}A(z) \\ &\geq c' \int_{|z|\geq R} |z|^{(1-m/2)(p-2)-2p} \, \mathrm{d}A(z). \end{aligned}$$

The last integral diverges, because $(1-m/2)(p-2)-2p+2 = m(1-p/2)-p \geq 0$ when $m \geq 2p/(2-p)$. By Lemma 3, $H_{\bar f} \notin S_p$. □

The proof is the same as the proof of [1, Theorem 1.6] via IMO, since $G(\bar f) = \mathrm{MO}(f)$ here; the only new point is that for $p > 1$ the divergence must come from ρ^{p-2}, which requires ρ to decay.

Part (c): oscillating strips on $F^2_{y^{2N+2}}$. For $p > 2$ the weight ρ^{p-2} is small where ρ is small, and we want the cut-offs of the symbol, where $\bar\partial f \neq 0$, to live there. For $\phi(x + iy) = y^{2N+2}$ the radius ρ is of order 1 near the real axis and decays like $|y|^{-N}$ away from it. The symbol consists of the bounded holomorphic functions $e^{i(z-x_k)/s_k}$, which oscillate with wavelength $2\pi s_k$ along the real axis, cut off to long rectangles of length L_k and height s_k. The derivative $1/s_k$ is felt along the real axis, where $\rho \simeq 1$, over a length L_k; the vertical cut-offs cost $\rho^{p-2} \simeq s_k^{-N(p-2)}$ per unit area, which is small.

We first record the geometry. Here $\Delta\phi = c_N y^{2N}$ with $c_N = (2N+2)(2N+1)$. For $z = x + iy$ and $s > 0$,

$$\mu(D(z,s)) \simeq s^2 \max(|y|, s)^{2N}.$$

Indeed, $|\operatorname{Im} w| \leq |y| + s$ on $D(z,s)$, which gives the upper bound. If $s \leq |y|/2$, then $|\operatorname{Im} w| \geq |y|/2$ on $D(z,s)$. If $s > |y|/2$, then $D(z,s)$ contains the disk of radius $s/4$ centred at $z + 3i\sigma s/4$, where $\sigma = \pm 1$ is the sign of y (or 1 if

$y = 0$), and on this disk $|\operatorname{Im} w| \geq s/2$. This gives the lower bound. Hence μ is doubling, and solving $\mu(D(z,\rho)) = 1$ gives

$$\rho(x+iy) \simeq (1+|y|)^{-N}.$$

In particular there are constants $C_1 \geq 1$ and $c_1 > 0$ with $\rho \leq C_1$ on $\mathbb{C}$, $\rho \geq c_1$ on $\{|y| \leq 2\}$, and $\rho^{p-2} \leq C_1 s^{-N(p-2)}$ on $\{|y| \geq s/2\}$ for $s \geq 1$.

Proof of Theorem 2 (c). Let $s_k = 2^k$ and $L_k = s_k^p = 2^{kp}$, and choose real numbers x_k such that the rectangles $Q_k = \{|x - x_k| \leq 2L_k + C_1,\ |y| \leq 2s_k + C_1\}$ are pairwise disjoint. Let $\chi \in C^\infty(\mathbb{R})$ satisfy $0 \leq \chi \leq 1$, $\chi = 1$ on $[-1,1]$ and $\chi = 0$ off $[-2,2]$. We consider $k \geq k_0$, where $s_{k_0} \geq 4C_1 + 4$, and put

$$f(x+iy) = \sum_{k \geq k_0} g_k(z)\chi\Big(\frac{x-x_k}{L_k}\Big)\chi\Big(\frac{y}{s_k}\Big), \qquad g_k(z) = e^{i(z-x_k)/s_k}.$$

The terms are smooth with disjoint supports, and $|g_k| = e^{-y/s_k} \leq e^2$ on the support of the kth term, so f is smooth and $|f| \leq e^2$.

$H_f \in S_p$. Since g_k is holomorphic, $\bar\partial f$ comes from the cut-offs only. On the support of the kth term we have $|\bar\partial f| \leq C/L_k$ where $|y| \leq s_k$, since there only the factor in x varies, and $|\bar\partial f| \leq C/s_k$ everywhere, since $L_k \geq s_k$. The disk $D^{1/2}(z)$ has radius at most $C_1/2$. Hence $G(f)(z) = 0$ unless z lies within $C_1/2$ of the support of $\bar\partial f$, and then $z \in Q_k$ for exactly one k. By Lemma 4 (a), $\rho^{-2}G(f)^p \leq \rho^{p-2}\sup_{D^{1/2}(z)}|\bar\partial f|^p$. We split Q_k into two parts. Where $|y| \leq s_k - C_1$, the disk $D^{1/2}(z)$ lies in $\{|y| < s_k\}$, so the supremum is at most C/L_k, and $\bar\partial f$ vanishes near z unless $L_k - C_1 \leq |x - x_k|$; there $\rho^{p-2} \leq C_1^{p-2}$. Where $|y| > s_k - C_1 \geq s_k/2$, the supremum is at most C/s_k and $\rho^{p-2} \leq C_1 s_k^{-N(p-2)}$. Both regions have area at most $CL_k s_k$, so

$$\begin{aligned}\int_{\mathbb{C}} \rho^{-2}G(f)^p \,\mathrm{d}A &\leq C\sum_k (L_k^{1-p}s_k + s_k^{1-p-N(p-2)}L_k)\\ &= C\sum_k (2^{k(1+p-p^2)} + 2^{k(1-N(p-2))}).\end{aligned}$$

Both exponents are negative, since $p^2 - p - 1 > 0$ for $p > 2$ and $N(p-2) > 1$. By Lemma 3, $H_f \in S_p$.

$H_{\bar f} \notin S_p$. Let $E_k = \{|x - x_k| \leq L_k - C_1,\ |y| \leq 1\}$. For $z \in E_k$ the closed disk $\overline{D^{1/2}(z)}$ lies in $\{|x - x_k| < L_k,\ |y| < s_k\}$, where $f = g_k$ is holomorphic. By Lemma 4 (b), $G(\bar f)(z) \geq \rho(z)|g_k'(z)|/(2\sqrt{2}) \geq c\,c_1/s_k$, because $|g_k'(z)| = e^{-y/s_k}/s_k \geq e^{-1}/s_k$. Since $\rho^{-2} \geq C_1^{-2}$ and $|E_k| \geq L_k$,

$$\int_{\mathbb{C}} \rho^{-2}G(\bar f)^p \,\mathrm{d}A \geq c\sum_k s_k^{-p}|E_k| \geq c\sum_k s_k^{-p}L_k = \infty.$$

By Lemma 3, $H_{\bar f} \notin S_p$. □

Part (a): poles on flat spots. For $0 < p \le 1$ we must deal with an arbitrary doubling weight. The idea is that of Xia's symbol: a pole $q/(w-z_0)$ cut off near z_0 has $\bar\partial f$ supported on one disk, while $|f'|^p = q^p|w-z_0|^{-2p}$ is not integrable at infinity when $p \le 1$. The difficulty is the weight ρ^{p-2}, which may decay where ρ grows. We therefore look for a region where ρ is essentially constant on a disk which is large compared with ρ itself. The following lemma shows that every doubling weight has such flat spots arbitrarily far out.

Lemma 5. *There is a constant $C_2 \ge 1$ with the following property. For all $R_0 > 1$ and $M > 1$ there are $z \in \mathbb{C}$ and $T > 0$ such that $|z| \ge R_0$, $M\rho(z) \le T \le |z|/2$, and $\rho \le C_2\rho(z)$ on $D(z,T)$.*

Proof. Fix $b > (\eta+\beta)/(1-\beta)$, with η and β as in (R3), and put $C_2 = 2^b$. Let $R \ge R_0$ and let c be a point with $|c| = 2R$. On the closed disk $\overline{D(c,R)}$, which lies in $\{R \le |w| \le 3R\}$, consider

$$Q(w) = \rho(w)d(w)^b, \qquad d(w) = R - |w-c|.$$

By (R1), Q is continuous. It vanishes on the boundary and is positive inside, so it attains its maximum at an interior point z. Put $d_* = d(z)$ and $T = d_*/2$; then $T \le R/2 \le |z|/2$. For $w \in D(z,T)$ we have $d(w) \ge d_*/2$, hence

$$\rho(w) \le \frac{Q(z)}{d(w)^b} \le 2^b\rho(z).$$

It remains to see that $T/\rho(z)$ is large. By (R3),

$$\rho(z)d_*^b = Q(z) \ge Q(c) \ge C_0^{-1}(2R)^{-\eta}R^b \qquad \text{and} \qquad \rho(z) \le C_0(3R)^\beta.$$

The two inequalities give $d_* \ge c_0R^{1-(\eta+\beta)/b}$, and hence

$$\frac{T}{\rho(z)} \ge c_0'R^{1-\beta-(\eta+\beta)/b},$$

with constants $c_0, c_0' > 0$ independent of R. The exponent is positive by the choice of b, so $T \ge M\rho(z)$ once R is large enough. □

The maximizer of $\rho\, d^b$ cannot sit where ρ increases steeply, and this is why ρ is nearly constant around it. The symbol for part (a) is a sum of cut-off poles placed at such flat spots z_k, with residues chosen so that the cut-off costs are summable while the gain on each flat spot is not.

Proof of Theorem 2 (a). Let $0 < p \le 1$, put $\varepsilon_k = k^{-2/p}$, and let C_2 be as in Lemma 5. We choose $z_1, z_2, \dots$ inductively by Lemma 5, together with radii T_k. Write $\rho_k = \rho(z_k)$, $\tau_k = e^{k^2}\rho_k$ and

$$q_k = \varepsilon_k\rho_k, \qquad S_{k-1} = \sum_{j<k} q_j, \qquad \delta_k = \frac{q_k}{16\tau_k^2}.$$

We require $T_k \ge \tau_k$ (take $M = e^{k^2}$), $|z_k| \ge 4|z_{k-1}|$, $|z_k| \ge 4^k$, and

$$64S_{k-1}\tau_k^2 \le q_k|z_k|^2, \qquad \frac{4q_k}{|z_k|^2} \le 2^{-k}\min_{j<k}\delta_j. \tag{6}$$

This is possible because $\tau_k^2/q_k = e^{2k^2}\rho_k/\varepsilon_k$ and $q_k \le \rho_k \le C_0|z_k|^\beta$ by (R3), so both conditions hold as soon as $|z_k|^{2-\beta}$ is large compared with quantities fixed before step k, and Lemma 5 allows $|z_k| \ge R_0$ for any R_0. Note also that $\rho_k \le e^{-k^2}T_k \le e^{-k^2}|z_k|/2$.

Let $\chi \in C^\infty(\mathbb{C})$ satisfy $0 \le \chi \le 1$, $\chi = 0$ on $D(0,1/2)$ and $\chi = 1$ off $D(0,1)$, put $\chi_k(w) = \chi((w-z_k)/\rho_k)$, and define

$$f(w) = \sum_{k\ge 1} \frac{q_k\chi_k(w)}{w-z_k}.$$

f is bounded and smooth. If $|w-z_k| < |z_k|/2$, then $|z_k|/2 < |w| < 3|z_k|/2$. Since $|z_k| \ge 4|z_{k-1}|$, these annuli are disjoint, so $|w-z_k| < |z_k|/2$ for at most one k. That term is at most $2q_k/\rho_k = 2\varepsilon_k \le 2$, because $\chi_k = 0$ where $|w-z_k| < \rho_k/2$. Every other term is at most $2q_j/|z_j| \le 2C_0|z_j|^{\beta-1} \le 2C_0 4^{j(\beta-1)}$. Hence f is bounded. On a compact set, all but finitely many terms are holomorphic (since $\chi_j = 1$ off $D(z_j,\rho_j)$) and bounded by $2C_0 4^{j(\beta-1)}$, so f is smooth, and f is holomorphic off $\bigcup_k D(z_k,\rho_k)$.

$H_f \in S_p$. We have $\bar\partial f = \sum_k q_k\bar\partial\chi_k/(w-z_k)$, which is supported in the disjoint annuli $B_k = \{\rho_k/2 \le |w-z_k| \le \rho_k\}$, and there $|\bar\partial f| \le 2\|\nabla\chi\|_\infty\varepsilon_k/\rho_k$. Suppose that $D^{1/2}(z)$ meets B_k at a point w. Then by (R1), $\rho(z) \le \rho(w) + \rho(z)/2$ and $\rho(w) \le \rho_k + |w-z_k| \le 2\rho_k$, so $\rho(z) \le 4\rho_k$ and $|z-z_k| \le 3\rho_k$. Since the disks $D(z_k,3\rho_k)$ are disjoint, $D^{1/2}(z)$ meets at most one B_k, and Lemma 4 (a) gives $G(f)(z) \le 8\|\nabla\chi\|_\infty\varepsilon_k$. Off $\bigcup_k D(z_k,3\rho_k)$ we have $G(f) = 0$. By (R2) with $s = 3$,

$$\int_{\mathbb{C}} \rho^{-2}G(f)^p \,\mathrm{d}A \le C\sum_k \varepsilon_k^p = C\sum_k k^{-2} < \infty,$$

and Lemma 3 gives $H_f \in S_p$.

$H_{\bar f} \notin S_p$. Let $A_k = \{(C_2+2)\rho_k \le |z-z_k| < \tau_k\}$. This annulus lies in $D(z_k,T_k)$, so $\rho \le C_2\rho_k$ on A_k, and it is here, on the flat spot, that we gain. Let $z \in A_k$, with k so large that $e^{k^2} \ge 4C_2$. The disk $D^{1/2}(z)$ has radius at most $C_2\rho_k/2$, so it stays at distance at least ρ_k from z_k. For $j < k$ we have $|z| \ge |z_k| - \tau_k \ge |z_k|/2$ and $|z_j| \le |z_k|/4$, so $|z-z_j| \ge |z_k|/4$, and points of $D^{1/2}(z)$ are at distance at least $|z_k|/4 - |z_k|/16 > |z_k|/8 \ge \rho_j$ from z_j, because $C_2\rho_k/2 \le |z_k|/16$. For $j > k$ we have $|z| \le 3|z_k|/2 \le 3|z_j|/8$, so $|z-z_j| \ge |z_j|/2$, and similarly $D^{1/2}(z)$ stays away from $D(z_j,\rho_j)$. Hence every χ_j equals 1 near $\overline{D^{1/2}(z)}$, so f is holomorphic there, and

$$f'(z) = -\sum_j \frac{q_j}{(z-z_j)^2}.$$

The term $j = k$ dominates. By the bounds just proved and (6),

$$\sum_{j\ne k} \frac{q_j}{|z-z_j|^2} \le \frac{16S_{k-1}}{|z_k|^2} + \sum_{j>k}\frac{4q_j}{|z_j|^2} \le \frac{q_k}{4\tau_k^2} + \delta_k \le \frac{q_k}{2|z-z_k|^2},$$

so $|f'(z)| \geq q_k/(2|z-z_k|^2)$. By Lemma 4 (b), and since $p-2<0$ and $\rho \leq C_2\rho_k$ on A_k,

$$\int_{A_k} \rho^{-2} G(\bar{f})^p \,\mathrm{d}A \geq c \int_{A_k} \rho^{p-2}|f'|^p \,\mathrm{d}A \geq c'\varepsilon_k^p \rho_k^{2p-2} \int_{(C_2+2)\rho_k}^{\tau_k} t^{1-2p}\,\mathrm{d}t$$
$$= c'\varepsilon_k^p \int_{C_2+2}^{e^{k^2}} u^{1-2p}\,\mathrm{d}u.$$

For $p=1$ the right-hand side is $c'k^{-2}(k^2-\log(C_2+2))$, which tends to c'. For $p<1$ it is at least $c''k^{-2}e^{(2-2p)k^2}$ for large k. The annuli A_k are pairwise disjoint, so $\int \rho^{-2}G(\bar{f})^p\,\mathrm{d}A=\infty$, and Lemma 3 gives $H_{\bar{f}} \notin S_p$. □

The proof shows where $p \leq 1$ is used: the gain on the kth flat spot is $\varepsilon_k^p \int_{C_2+2}^{e^{k^2}} u^{1-2p}\,\mathrm{d}u$, which is unbounded in k only when $p \leq 1$. For $p>1$ the integral stays bounded and the gains are summable. A counterexample for $p>1$ must therefore exploit the weight ρ^{p-2} itself, as in parts (b) and (c), and when $\Delta\phi \simeq 1$ there is none by [7, Theorem 3].

5. Compact Hankel operators on doubling Fock spaces

The proof of Theorem 3 has three steps. Compactness of H_f gives $G_r(f)(z) \to 0$ as $z \to \infty$, by testing on normalized reproducing kernels. A rescaling argument, which is where boundedness of f and Liouville's theorem enter, upgrades this to $\mathrm{MO}_s(f)(z) \to 0$. Finally the characterization of Chen and Xu of simultaneous compactness gives that H_f and $H_{\bar{f}}$ are both compact. The key step is the second. To see why it should hold, note that on a disk of radius comparable to $\rho(z)$, an f with small $G_r(f)(z)$ is close to a holomorphic function h. If this held for all disks at all scales, h would be a bounded entire function, hence constant. Rescaling by $\rho(z)$, which is $o(|z|)$, produces such an entire limit at infinity.

The same mechanism drives the proof of [8, Theorem 1.2] for weights with real Hessian comparable to the identity. There the local holomorphic approximants on balls of a large fixed radius R are bounded, so by the Cauchy estimates their derivatives are $O(1/R)$, and this requires $G_R(f)(z) \to 0$ for large R. Testing on kernels gives $G_r(f)(z) \to 0$ only for $r \leq r_0$, and on a doubling Fock space we pass to large scales by rescaling and normal families instead.

Lemma 6. *Suppose that $f \in L^\infty$ and that $G_r(f)(z) \to 0$ as $z \to \infty$ for some $r>0$. Then $\mathrm{MO}_s(f)(z) \to 0$ as $z \to \infty$ for every $s>0$.*

Proof. It suffices to show that every sequence $z_n \to \infty$ has a subsequence along which $\mathrm{MO}_s(f)(z_n) \to 0$. Put $\rho_n = \rho(z_n)$ and $f_n(\zeta) = f(z_n + \rho_n\zeta)$, so that $|f_n| \leq \|f\|_\infty$ and

$$\mathrm{MO}_s(f)(z_n)^2 \leq \frac{1}{\pi s^2} \int_{D(0,s)} |f_n - c|^2 \,\mathrm{d}A$$

for every constant c.

Local holomorphic approximation. Fix $R > 0$ and let $|\zeta| \le R$. The point $w = z_n + \rho_n \zeta$ lies in $D^{R+1}(z_n)$, so $\rho(w) \ge c_{R+1}^{-1} \rho_n$ by (R2). Put $\kappa_R = r/c_{R+1}$. Then $D(w, \kappa_R \rho_n) \subset D^r(w)$, and there is $h \in H(D^r(w))$ with normalized L^2 distance to f on $D^r(w)$ at most $G_r(f)(w) + 1/n$. Changing variables, the function $h_n(\xi) = h(z_n + \rho_n \xi)$ is holomorphic on $D(\zeta, \kappa_R)$ and

$$\int_{D(\zeta,\kappa_R)} |f_n - h_n|^2 \, \mathrm{d}A \le \pi r^2 c_{R+1}^2 \big(G_r(f)(w) + 1/n\big)^2.$$

Since $\rho_n/|z_n| \to 0$ by (R3), we have $|w| \ge |z_n| - (R+1)\rho_n \to \infty$ uniformly in $|\zeta| \le R$, so the right-hand side tends to 0 uniformly in $|\zeta| \le R$.

Normal families. The functions f_n are bounded by $\|f\|_\infty$, so a subsequence converges weak-star in $L^\infty(\mathbb{C})$ to a function F with $|F| \le \|f\|_\infty$. Fix ζ and $R > |\zeta|$, and let h_n be as above. The functions h_n are bounded in $L^2(D(\zeta, \kappa_R))$, so by the mean value inequality they are uniformly bounded on $D(\zeta, \kappa_R/2)$, and Montel's theorem gives a further subsequence converging uniformly on $D(\zeta, \kappa_R/4)$ to a holomorphic function h_ζ. Along it, $f_n \to h_\zeta$ in $L^2(D(\zeta, \kappa_R/4))$. Weak-star limits are unique, so $F = h_\zeta$ almost everywhere on $D(\zeta, \kappa_R/4)$. Since the limit does not depend on the further subsequence, the whole weak-star convergent subsequence converges to F in $L^2(D(\zeta, \kappa_R/4))$. Thus F coincides almost everywhere with a function which is holomorphic near every point, that is, with an entire function. It is bounded by $\|f\|_\infty$, so it is a constant c by Liouville's theorem.

Conclusion. Covering $\overline{D(0,s)}$ by finitely many of the disks $D(\zeta, \kappa_R/4)$ with $R = s + 1$, we find $f_n \to c$ in $L^2(D(0,s))$ along the subsequence. By the inequality at the beginning of the proof, $\mathrm{MO}_s(f)(z_n) \to 0$ along this subsequence. □

We also need the following kernel estimates from [1, Lemma 2.3], which collects results of Marzo and Ortega-Cerdà [10] and others. Let ϕ be smooth, let K_z be the reproducing kernel of F^2_ϕ and $k_z = K_z/\|K_z\|_{2,\phi}$. Then $k_z \to 0$ uniformly on compact sets as $z \to \infty$ [1, Lemma 2.3 (3)], $\|K_z\|_{2,\phi} \simeq e^{\phi(z)} \rho(z)^{-1}$ [1, Lemma 2.3 (4)], and there is $r_0 > 0$ such that $|K_z(w)| \simeq e^{\phi(w)+\phi(z)} \rho(z)^{-2}$ for $w \in D^{r_0}(z)$ [1, Lemma 2.3 (2)]. Together these say that

$$|k_z(w)| e^{-\phi(w)} \simeq \rho(z)^{-1}, \qquad w \in D^{r_0}(z). \tag{7}$$

Finally, we use the case $p = q = 2$ of a theorem of Chen and Xu [4, Theorem 3.4], which characterizes simultaneous compactness of H_f and $H_{\bar f}$ on doubling Fock spaces by vanishing mean oscillation. Chen and Xu work with an arbitrary subharmonic $\phi \not\equiv 0$ such that $\Delta\phi \, \mathrm{d}A$ is doubling, and for $1 \le p \le q < \infty$ they prove that $H_f, H_{\bar f} \colon F^p_\phi \to L^q_\phi$ are both compact if and only if f lies in $\mathrm{VMO}^{2(1/q-1/p),q}_r$. For $p = q = 2$ the weight exponent is 0, their mean oscillation $\mathrm{MO}_{2,r}$ is our MO_r, and for bounded f the condition is $\mathrm{MO}_r(f)(z) \to 0$ as $z \to \infty$. Hence, if $f \in L^\infty$ and $\mathrm{MO}_r(f)(z) \to 0$ as $z \to \infty$ for every $r > 0$, then H_f and $H_{\bar f}$ are both compact on F^2_ϕ.

Proof of Theorem 3. By Lemma 3 (a) and the symmetry of the statement, we may assume that ϕ is smooth and that H_f is compact; we show that $H_{\bar f}$ is compact.

We first show that $k_z \to 0$ weakly as $z \to \infty$. The functions k_z have norm 1, and for every w we have $\langle k_z, K_w\rangle = k_z(w) \to 0$. Since the kernels K_w span a dense subspace of F^2_ϕ, this gives weak convergence. As H_f is compact, $\|H_f k_z\|_{2,\phi} \to 0$.

Next we bound $\|H_f k_z\|$ from below by $G_{r_0}(f)(z)$. By (7), k_z has no zeros in $D^{r_0}(z)$, so $h_z = P(fk_z)/k_z$ is holomorphic there. Since $H_f k_z = (f - h_z)k_z$ on $D^{r_0}(z)$, (7) gives

$$\begin{aligned}\|H_f k_z\|^2_{2,\phi} &\geq \int_{D^{r_0}(z)} |f - h_z|^2 |k_z|^2 e^{-2\phi}\,\mathrm{d}A \\ &\geq \frac{c}{\rho(z)^2}\int_{D^{r_0}(z)} |f - h_z|^2\,\mathrm{d}A \geq c\pi r_0^2\, G_{r_0}(f)(z)^2.\end{aligned}$$

Hence $G_{r_0}(f)(z) \to 0$ as $z \to \infty$. By Lemma 6, $\mathrm{MO}_s(f)(z) \to 0$ for every $s > 0$, and the theorem of Chen and Xu shows that $H_{\bar f}$ is compact. □

6. Concluding remarks

The set of exponents for a given weight. Corollary 1 answers Open Problem B in the form in which it is introduced in [7] and restated in [1]. Read literally, the question asks for the set

$$\mathcal{P}_\phi = \{p \in (0,\infty) :\ \text{the BCP holds for } S_p(F^2_\phi, L^2_\phi)\}$$

for each individual weight ϕ. Our results show that $\mathcal{P}_\phi \subset (1,\infty)$ and $2 \in \mathcal{P}_\phi$ for every ϕ, that $\mathcal{P}_\phi = (1,\infty)$ when $\Delta\phi \simeq 1$, that $p \notin \mathcal{P}_\phi$ for $\phi(z) = |z|^m$ and $1 < p < 2$, $m \geq 2p/(2-p)$, and that $p \notin \mathcal{P}_\phi$ for $\phi(x+iy) = y^{2N+2}$ and $p > 2 + 1/N$. We do not know $\mathcal{P}_\phi$ for these two weights, for instance whether $\mathcal{P}_{|z|^m}$ contains all $p > 2$.

Problem 1. *Let ϕ be subharmonic with $\Delta\phi\,\mathrm{d}A$ doubling. Determine $\mathcal{P}_\phi$ in terms of ρ. Is $\mathcal{P}_\phi$ always an interval?*

Muckenhoupt weights. As noted in [1, Remark 6.1], for $1 < p < \infty$ the approach of Asghari, Hu and Virtanen would give the BCP for S_p if $\omega = \rho^{p-2}$ were a Muckenhoupt A_p weight, through the boundedness of the Beurling–Ahlfors transform on $L^p(\omega)$. Our counterexamples are consistent with this. Recall that $(1+|z|)^a \in A_p$ on $\mathbb{C}$ if and only if $-2 < a < 2(p-1)$, and that $(1+|y|)^{-b} \in A_p$ on $\mathbb{C}$ if and only if $-(p-1) < b < 1$; the failure at the upper endpoints is seen by testing [1, (6.26)] on the disks $D(0,s)$ with $s \to \infty$. For $\phi(z) = |z|^m$ and $1 < p < 2$ we have $\rho^{p-2} \simeq (1+|z|)^{(m/2-1)(2-p)}$, by [1, Lemma 2.5] and the continuity of ρ, and this weight fails to be in A_p exactly when $(m/2-1)(2-p) \geq 2(p-1)$, that is, when $m \geq 2p/(2-p)$. For $\phi(x+iy) = y^{2N+2}$ and $p > 2$ the weight $\rho^{p-2} \simeq (1+|y|)^{-N(p-2)}$ fails to be in A_p exactly when $N(p-2) \geq 1$. Thus the conditions in parts (b) and (c)

of Theorem 2 are, up to the endpoint $N(p-2)=1$ in (c), the conditions under which $\rho^{p-2}\notin A_p$.

Problem 2. *Let $1<p<\infty$, $p\neq 2$. Is it true that the BCP holds for $S_p(F^2_\phi,L^2_\phi)$ if and only if ρ^{p-2} is an A_p weight?*

Unbounded symbols. The fourth open problem of [7], the characterization of unbounded symbols for which the BCP holds on Fock spaces, is not addressed here. For compact operators, the proof of Theorem 3 uses boundedness of f only in the Liouville step of Lemma 6, so a growth condition on the local holomorphic approximants would suffice there; this gives sufficient conditions only.

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